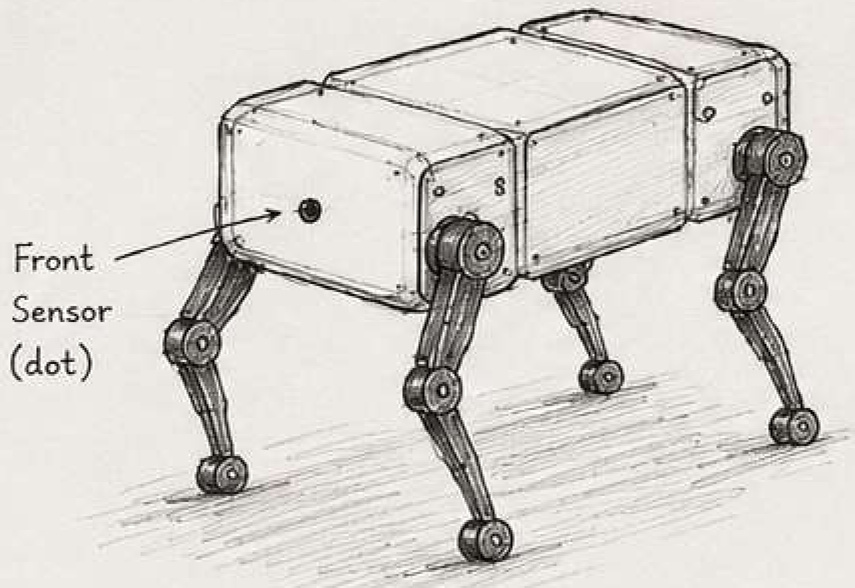
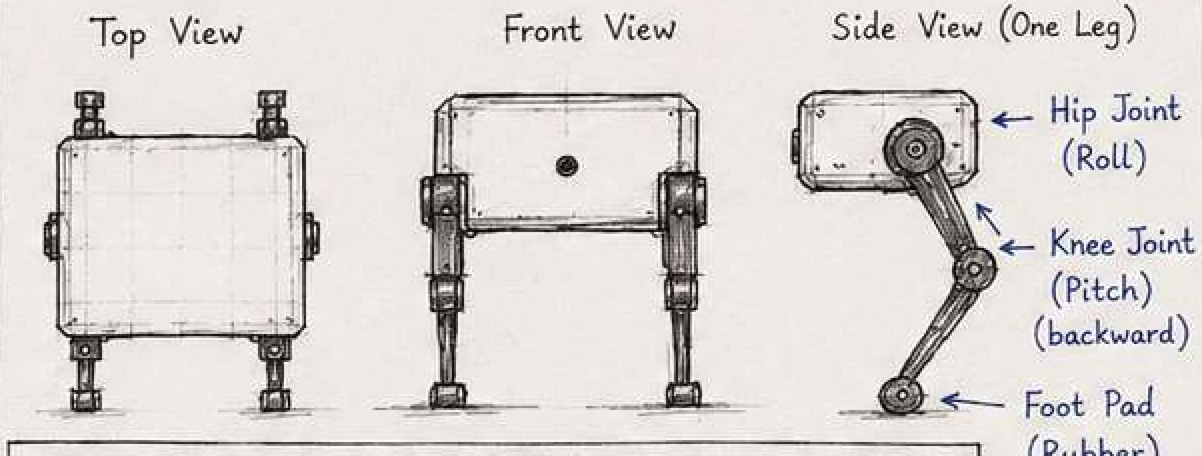
1) SKETCH (Hand-drawn style)



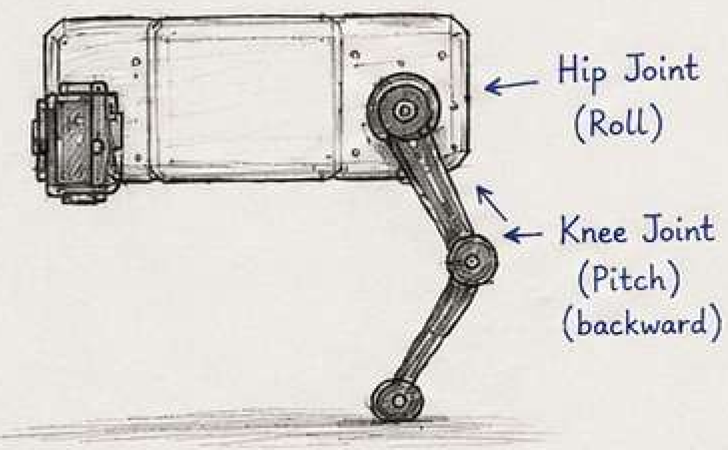
2) BODY & LEG DESIGN



Body Shape	:	Compact rectangular body with rounded edges.
Front Sensor	:	Single front sensor (dot) in the middle.
Material	:	3D Printed Parts (PLA/ABS)
Frame	:	Aluminum / Carbon Fiber (Lightweight)
Leg Type	:	2-DOF per leg (Hip + Knee) with reverse knee
Foot	:	Rubber pad (rounded) for better grip

3) JOINTS & DEGREES OF FREEDOM

- Each Leg : 2 Joints (2-DOF)
- Total : 8 Joints for 4 Legs
- DOF = 8



4) MOTORS SELECTION

Each Joint ⟶ 1x DC Servo Motor (High Torque)

Recommended Motor :

Dynamixel XM430-W210

- Stall Torque : 2.1 Nm @ 11.1V
- Weight : 64 g
- Control : TTL Serial
- Built-in Encoder

DYNAMIXEL
XM430-W210

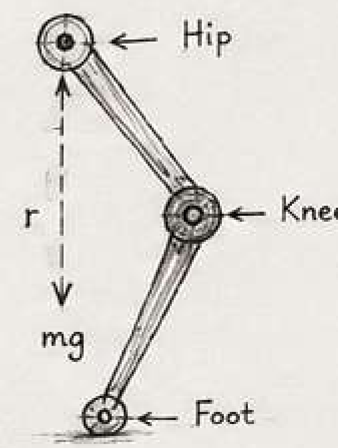
5) INITIAL TORQUE CALCULATION (Example: Knee Joint)

Assume :

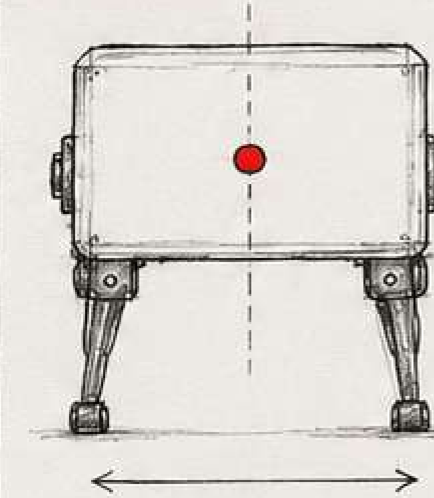
- Mass of robot (m) : 2.5 kg
- Each leg supports ~25% ⟶ 0.625 kg
- Distance from knee to foot (r) : 0.10 m
- Gravity (g) : 9.81 m/s²

$$\text{Torque } (\tau) = \text{Force} \times \text{Distance} = (0.625 \times 9.81) \times 0.10 = 0.613 \text{ Nm}$$

Choose a motor with torque $\geq$ 0.613 Nm
(Safety factor 2x ⟶ $\geq$ 1.2 Nm)



6) STABILITY & CENTER OF GRAVITY

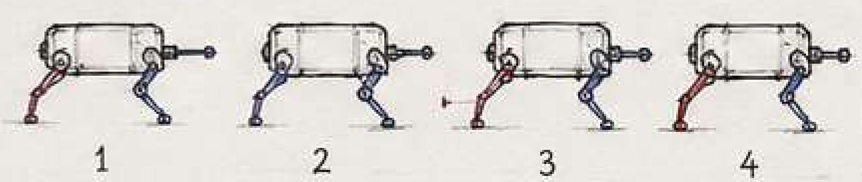


- Keep CoG (Center of Gravity) in the middle and as low as possible.
- Wide stance for better stability.

7) PROPOSED GAIT (DIAGONAL TROT)

Diagonal legs move together :

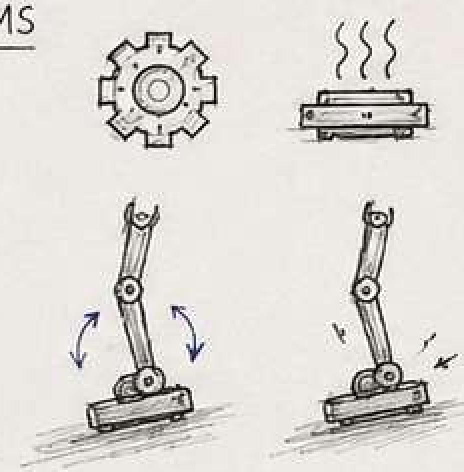
- Front Left + Rear Right
- Front Right + Rear Left



Red : Moving Leg
Blue : Supporting Leg (Stance)

8) EXPECTED MECHANICAL PROBLEMS

- Joint backlash
- Motor overheating
- Leg misalignment
- Slippage on smooth surfaces
- Structural stress on joints
- Battery weight affecting balance
- Wear and tear on foot pads



9) NOTES

- Simple, lightweight and efficient design.
- Suitable for indoor/outdoor flat terrain.
- Easy to maintain and modify.

Milaf Ali Aljuaid